

Canonical Cover

{ID} -> {as_of_year, respondent_id, agency_code, loan_type, property_type, loan_purpose, owner_occupancy, preapproval, action_taken, msamd, state_code, county_code, census_tract_number, applicant_ethnicity, co_applicant_ethnicity, applicant_race_1, applicant_race_2, applicant_race_3, applicant_race_4, applicant_race_5, co_applicant_race_1, co_applicant_race_2, co_applicant_race_3, co_applicant_race_4, co_applicant_race_5, applicant_sex, co_applicant_sex, purchaser_type, denial_reason_1, denial_reason_2, denial_reason_3, loan_amount_000s, applicant_income_000s, rate_spread, sequence_number, application_date_indicator, hoepa_status, lien_status, edit_status}

{agency_code} -> {agency_name, agency_abbr}

{loan_type} -> {loan_type_name}
{property_type} -> {property_type_name}
{loan_purpose} -> {loan_purpose_name}
{owner_occupancy} -> {owner_occupancy_name}
{preapproval} -> {preapproval_name}
{action_taken} -> {action_taken_name}
{purchaser_type} -> {purchaser_type_name}
{hoepa_status} -> {hoepa_status_name}
{lien_status} -> {lien_status_name}
{edit_status} -> {edit_status_name}

{msamd} -> {msamd_name}

{state_code} -> {state_name, state_abbr}

{county_code} -> {county_name}

{applicant_ethnicity} -> {applicant_ethnicity_name}
{co_applicant_ethnicity} -> {co_applicant_ethnicity_name}

{applicant_race_1} -> {applicant_race_name_1}
{applicant_race_2} -> {applicant_race_name_2}
{applicant_race_3} -> {applicant_race_name_3}
{applicant_race_4} -> {applicant_race_name_4}
{applicant_race_5} -> {applicant_race_name_5}

{co_applicant_race_1} -> {co_applicant_race_name_1}
{co_applicant_race_2} -> {co_applicant_race_name_2}
{co_applicant_race_3} -> {co_applicant_race_name_3}
{co_applicant_race_4} -> {co_applicant_race_name_4}

{co_applicant_race_5} -> {co_applicant_race_name_5}

{applicant_sex} -> {applicant_sex_name}

{co_applicant_sex} -> {co_applicant_sex_name}

{denial_reason_1} -> {denial_reason_name_1}

{denial_reason_2} -> {denial_reason_name_2}

{denial_reason_3} -> {denial_reason_name_3}

{census_tract_number} -> {population, minority_population, hud_median_family_income,
tract_to_msamd_income, number_of_owner_occupied_units, number_of_1_to_4_family_units}

BCNF

LoanApplication(ID, as_of_year, respondent_id, agency_code, loan_type, property_type,
loan_purpose, owner_occupancy, preapproval, action_taken, msamd, state_code, county_code,
census_tract_number, applicant_ethnicity, co_applicant_ethnicity, applicant_race_1,
applicant_race_2, applicant_race_3, applicant_race_4, applicant_race_5, co_applicant_race_1,
co_applicant_race_2, co_applicant_race_3, co_applicant_race_4, co_applicant_race_5,
applicant_sex, co_applicant_sex, purchaser_type, denial_reason_1, denial_reason_2,
denial_reason_3, loan_amount_000s, applicant_income_000s, rate_spread, sequence_number,
application_date_indicator, hoepa_status, lien_status, edit_status)

Agency(agency_code, agency_name, agency_abbr)

LoanType(loan_type, loan_type_name)

PropertyType(property_type, property_type_name)

LoanPurpose(loan_purpose, loan_purpose_name)

OwnerOccupancy(owner_occupancy, owner_occupancy_name)

Preapproval(preapproval, preapproval_name)

ActionTaken(action_taken, action_taken_name)

PurchaserType(purchaser_type, purchaser_type_name)

HOEPASStatus(hoepa_status, hoepa_status_name)

LienStatus(lien_status, lien_status_name)

EditStatus(edit_status, edit_status_name)

MSAMD(msamd, msamd_name)

State(state_code, state_name, state_abbr)

County(county_code, county_name)

ApplicantEthnicity(applicant_ethnicity, applicant_ethnicity_name)

CoApplicantEthnicity(co_applicant_ethnicity, co_applicant_ethnicity_name)

ApplicantRace1(applicant_race_1, applicant_race_name_1)

ApplicantRace2(applicant_race_2, applicant_race_name_2)

ApplicantRace3(applicant_race_3, applicant_race_name_3)

ApplicantRace4(applicant_race_4, applicant_race_name_4)

ApplicantRace5(applicant_race_5, applicant_race_name_5)

CoApplicantRace1(co_applicant_race_1, co_applicant_race_name_1)
CoApplicantRace2(co_applicant_race_2, co_applicant_race_name_2)
CoApplicantRace3(co_applicant_race_3, co_applicant_race_name_3)
CoApplicantRace4(co_applicant_race_4, co_applicant_race_name_4)
CoApplicantRace5(co_applicant_race_5, co_applicant_race_name_5)
ApplicantSex(applicant_sex, applicant_sex_name)
CoApplicantSex(co_applicant_sex, co_applicant_sex_name)
DenialReason1(denial_reason_1, denial_reason_name_1)
DenialReason2(denial_reason_2, denial_reason_name_2)
DenialReason3(denial_reason_3, denial_reason_name_3)
CensusTract(census_tract_number, population, minority_population,
hud_median_family_income, tract_to_msamd_income, number_of_owner_occupied_units,
number_of_1_to_4_family_units)

List of all attributes total

ID
as_of_year,
respondent_id,
agency_name,
agency_abbr,
agency_code,
loan_type_name,
loan_type,
property_type_name,
property_type,
loan_purpose_name,
loan_purpose,
owner_occupancy_name,
owner_occupancy,
loan_amount_000s,
preapproval_name,
preapproval,
action_taken_name,
action_taken,
msamd_name,
msamd,
state_name,
state_abbr,
state_code,
county_name,
county_code,
census_tract_number,
applicant_ethnicity_name,
applicant_ethnicity,

co_applicant_ethnicity_name,
co_applicant_ethnicity,
applicant_race_name_1,
applicant_race_1,
applicant_race_name_2,
applicant_race_2,
applicant_race_name_3,
applicant_race_3,
applicant_race_name_4,
applicant_race_4,
applicant_race_name_5,
applicant_race_5,
co_applicant_race_name_1,
co_applicant_race_1,
co_applicant_race_name_2,
co_applicant_race_2,
co_applicant_race_name_3,
co_applicant_race_3,
co_applicant_race_name_4,
co_applicant_race_4,
co_applicant_race_name_5,
co_applicant_race_5,
applicant_sex_name,
applicant_sex,
co_applicant_sex_name,
co_applicant_sex,
applicant_income_000s,
purchaser_type_name,
purchaser_type,
denial_reason_name_1,
denial_reason_1,
denial_reason_name_2,
denial_reason_2,
denial_reason_name_3,
denial_reason_3,
rate_spread,
hoepa_status_name,
hoepa_status,
lien_status_name,
lien_status,
edit_status_name,
edit_status,
sequence_number,
population,

minority_population,
hud_median_family_income,
tract_to_msamd_income,
number_of_owner_occupied_units,
number_of_1_to_4_family_units,
application_date_indicator